

# REQUIREMENT ANALYSIS

## Proposed System

The proposed Smart Visitor Access & Security Management System is a centralized ServiceNow-based application designed to automate and streamline the visitor management process within an organization.

The system enables employees to register visitors before their arrival, validates visitor information, prevents duplicate registrations, verifies visitor eligibility through blacklist checks, automates managerial approvals, notifies security personnel, tracks visitor movement, and generates real-time analytical reports.

By replacing manual visitor registers and email-based approvals, the system improves workplace security, operational efficiency, compliance, and visitor experience.

## Proposed Workflow

Employee



Visitor Registration



Phone Number Validation



Visit Date Validation



Duplicate Visitor Verification



Blacklist Validation



Manager Approval



Security Notification



VIP Visitor Evaluation



Visitor Check-In



Visitor Check-Out



Auto Expiry Process



Reports & Dashboard Analytics

## **Key Features**

### **Digital Visitor Registration**

Employees can register visitors through a centralized ServiceNow application.

### **Automated Data Validation**

The system validates phone numbers and visit dates before submission.

### **Auto Generated Visitor ID**

Every visitor request is assigned a unique identification number automatically.

### **Duplicate Visitor Prevention**

The system prevents multiple registrations of the same visitor for the same visit date.

### **Blacklisted Visitor Detection**

Blacklisted visitors are automatically identified and restricted from receiving approval.

### **Approval Workflow Automation**

Visitor requests are routed automatically to the responsible manager for approval.

### **Security Notification System**

Approved visitor details are automatically shared with the security team.

### **Check-In / Check-Out Tracking**

Security personnel can record visitor entry and exit activities.

### **VIP Visitor Management**

VIP visitors receive high-priority handling and special security notifications.

### **Auto Expiry Management**

Expired visitor records are automatically identified and updated.

### **Reporting & Dashboard Analytics**

Management can monitor visitor activities through interactive reports and dashboards.

## **Functional Requirements**

Functional requirements define the actions and operations that the system must perform.

### **Visitor Registration Management**

The system shall allow employees to create visitor requests before the visitor arrives at the organization.

#### **Inputs**

- Visitor Name
- Phone Number
- Email Address
- Company Name
- Purpose of Visit
- Visit Date
- Host Employee
- VIP Visitor Indicator

#### **Data Validation**

The system shall validate visitor information before storing records.

#### **Validation Rules**

- Phone number must contain exactly ten digits.
- Only numeric values are allowed in the phone number field.
- Past visit dates shall not be accepted.
- Mandatory fields must be completed before submission.

#### **Visitor ID Generation**

The system shall automatically generate a unique Visitor ID for every visitor request.

#### **Example**

VIS0001

VIS0002

VIS0003

The generated ID shall be unique and used for visitor tracking and auditing purposes.

#### **Visitor Request Storage**

The system shall store all visitor records in a centralized Visitor Request table.

The stored information shall be available for reporting, auditing, and operational monitoring.

#### **Duplicate Visitor Detection**

The system shall identify duplicate visitor requests based on:

- Phone Number
- Visit Date

If a duplicate record exists, the system shall prevent submission and display an appropriate error message.

## **Blacklisted Visitor Verification**

The system shall verify visitor details against the Blacklisted Visitor database.

If a match is identified:

- The request shall be automatically rejected.
- Security personnel shall be informed.
- Access shall not be granted.

## **Approval Workflow**

The system shall automatically initiate an approval workflow when a visitor request is submitted.

The approval request shall be routed to the appropriate manager.

## **Approval Decision Management**

The manager shall have the ability to:

- Approve Visitor Request
- Reject Visitor Request

The system shall update the visitor status based on the approval decision.

## **Security Notification Management**

Upon successful approval, the system shall automatically send an email notification to the security team.

The notification shall contain:

- Visitor Name
- Visit Date
- Purpose of Visit
- Host Employee
- Approval Status

## **VIP Visitor Management**

The system shall support VIP visitor identification.

When a visitor is marked as VIP:

- Priority shall automatically be set to High.
- Security personnel shall receive a special VIP notification.
- The visitor shall receive prioritized processing.

## **Visitor Check-In Management**

The system shall allow security personnel to check in approved visitors.

During check-in:

- Status shall be updated to Checked In.
- Check-In Time shall be recorded automatically.

## **Visitor Check-Out Management**

The system shall allow security personnel to check out visitors.

During check-out:

- Status shall be updated to Checked Out.
- Check-Out Time shall be recorded automatically.

## **Auto Expiry Management**

The system shall automatically identify outdated visitor records through a Scheduled Job.

Conditions:

- Visit Date is earlier than the current date.
- Status is Approved.

The system shall automatically update the status to Expired.

## **Reporting and Analytics**

The system shall provide real-time reporting capabilities.

Reports shall include:

- Today's Visitors
- Approved Visitors
- Rejected Visitors
- Checked-In Visitors
- Checked-Out Visitors
- VIP Visitors
- Expired Visitors

## **Dashboard Monitoring**

The system shall provide a centralized dashboard for operational monitoring.

The dashboard shall display:

- Total Visitor Statistics
- Approval Status Summary
- VIP Visitor Information
- Check-In / Check-Out Metrics
- Expired Visitor Records

## Expected Outcome

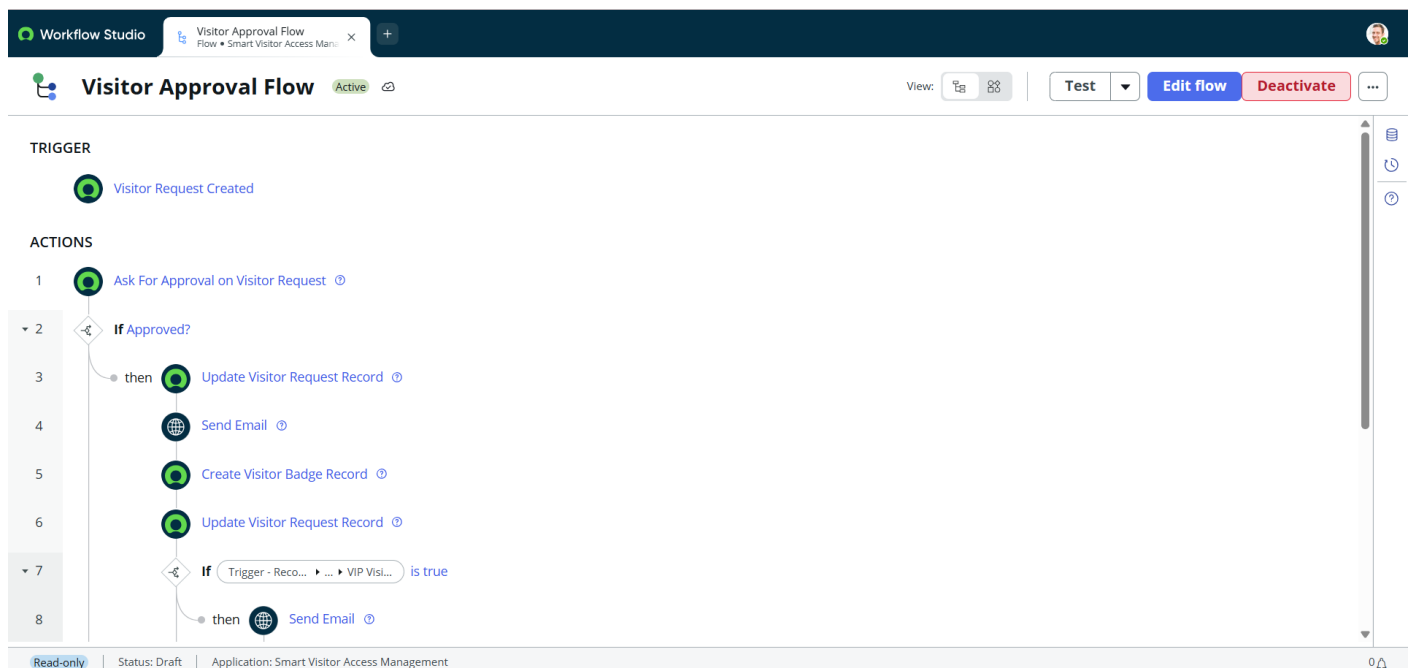
The Smart Visitor Access & Security Management System will provide a secure, automated, and scalable visitor management solution.

The implementation will:

- Improve workplace security
- Reduce manual administrative effort
- Accelerate visitor approval processes
- Prevent unauthorized visitor access
- Improve visitor tracking accuracy
- Enhance compliance and auditing capabilities
- Provide real-time operational visibility
- Support data-driven decision making through analytics and dashboards

The solution ensures an efficient, transparent, and secure visitor management process for modern organizations.

## Visitor Approval Workflow using Flow Designer



The Visitor Approval Workflow is implemented using ServiceNow Flow Designer.

When a visitor request is created, the system automatically initiates a manager approval process. Upon approval, the visitor status is updated, security notifications are sent, visitor badge records are generated, and VIP visitors trigger additional priority notifications.